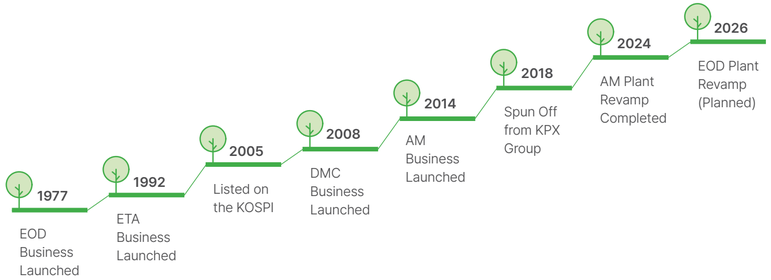
Green Chemical

Your Trusted Global Partner

Green 森 Chemical

HISTORY



ABOUT

Green Chemical is one of Korea's leading fine chemical companies, dedicated to providing innovative and reliable chemical solutions that meet the evolving needs of our global customers.

Our main manufacturing facility is located in the Daesan Petrochemical Complex on the west coast of Korea. Since 1977, we have been producing Ethylene Oxide Derivatives (EOD), building extensive expertise in EO-based chemistry. In 1992, Green Chemical successfully achieved Korea's first commercial production of Ethanolamines (ETA), further strengthening our position in the specialty chemical industry.

In 2008, we expanded our product portfolio by launching Dimethyl Carbonate (DMC), a versatile chemical used as an intermediate for polycarbonate (PC), a VOC-free solvent, and an electrolyte solvent for secondary batteries.

In 2014, Green Chemical further expanded into the specialty coatings market by developing UV-curable Acrylate Monomers (AM). In particular, we have developed proprietary hybrid monomers designed to enhance hardness and deliver a high refractive index for advanced coating applications.

Customer satisfaction remains our highest priority. Green Chemical is committed to delivering reliable products, technical expertise, and sustainable solutions to support the success of our partners worldwide.

CORE COMPETENCIES

1 EO & PO-based Specialty Chemical Platform

- Direct EO pipeline supply
- Advanced EO/PO derivative synthesis capability
- Extensive portfolio of specialty chemical applications

2 Advanced UV-Curable Acrylate Monomer Platform

- High-performance functional monomer development
- Hybrid monomer design for enhanced hardness and high refractive index
- Applications in advanced coatings for optical, display, and protective materials

3 Flexible, Rapid, and Customized Production System

- Specialized in multi-product and small-batch production
- Responsive to diverse customer formulation requirements
- Balancing high performance with sustainability

4 Global Network & Reliability

- Global supply network
- ISO 9001 & 14001 Certified
- Over 45 years of EO/PO chemistry expertise

PRODUCT PORTFOLIO

EOD (Ethylene Oxide Derivatives)

- Production Capacity: 45,000 MT/year
- Expanding Capacity to 70,000 MT/year by 2026
- Direct EO pipeline within Daesan Complex
- 14 multi-purpose batch reactors (3-30MT)

ETA (Ethanolamine)

- Production Capacity: 25,000 MT/year
- Korea's Sole domestic manufacturer
- Semiconductor process cleaning, industrial gas purification & more
- Stable domestic supply chain

AM (Acrylate Monomer)

- Production Capacity: 20,000 MT/year
- UV-curable functional monomers
- Advanced coating materials for diverse applications
- 11 multi-purpose batch reactors (6-30MT)

DMC (Dimethyl Carbonate)

- Production Capacity: 100,000 MT/year
- Key raw material for Polycarbonate (PC)
- Lithium-ion battery electrolyte
- VOC-free solvent alternative

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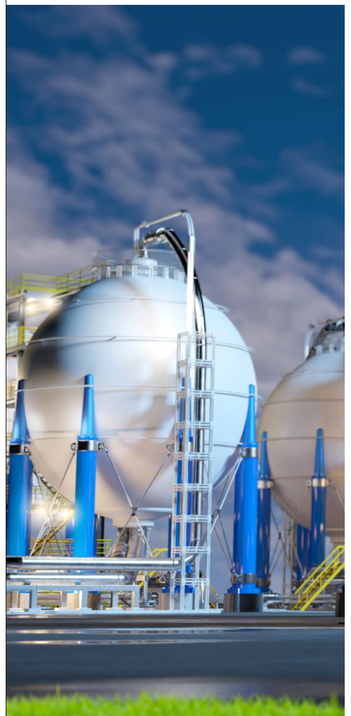


Why Choose Green Chemical

1977

Over 45 years of accumulated EO/ PO and derivative technology expertise

25

25 multi-purpose reactors (3-30MT) for flexible production (EOD & AM)

215K

Annual production of 215,000MT (EOD, ETA, DMC, AM)

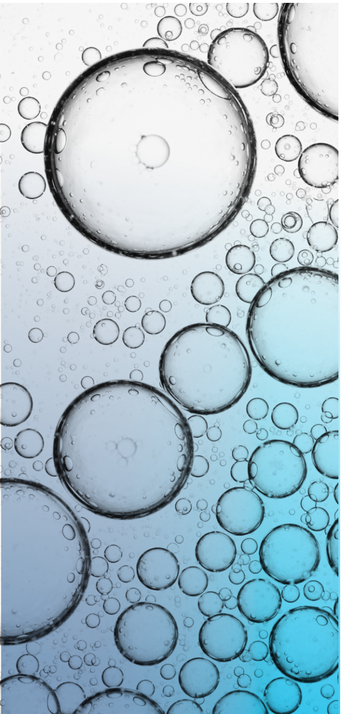
$230M

Annual Revenue

200+

Professional Workforce

EOD / SEGMENT 1



Industrial & Institutional (I&I) Cleaning and Water Treatment

Providing low-foam and defoaming agents, along with NPE-Free detergents for industrial applications.

Low-Foam agents (LF Series)

- Provides low-foam and effective foam control with excellent rinse performance and outstanding wetting properties

NPE-Free Eco-Friendly Detergents (NF, DA, TDA Series)

- Flagship environmentally friendly cleaners compliant with global regulations properties

Foam Control Copolymers (RP & PE Series)

- Strong defoaming/low-foaming performance
- Simultaneously provides excellent cleaning power

Specialty Polymers (TE Series)

- Amine-based tetra functional block copolymers
- Specialized for membrane cleaning, high-temperature Clean-in-Place

EOD / SEGMENT 2

Coating & High Performance Resin Intermediates

Key intermediates and high-value polymer solutions to maximize the performance of paints, coatings, and resins.

EOD for AM Production

- Products: BA, TM, PENTA, PH series, etc.
- Supplied to leading global coating and resin manufacturers
- Key raw materials for UV/thermosetting coatings

High purity Polyalkylene Glycol

- Catalyst-Free PEG : SK series optimized for high-performance spandex & polyurethane synthesis
- Advanced EO/PO Copolymers
  High-quality RT & FC series tailored for diverse industrial needs

EOD / SEGMENT 3

Agrochemical Formulation Agents

We provide to enhance the stability, penetration, and spreading ability of agrochemical agents.

Premium Dispersants (SP, DS, TSP Series)

- Exceptional suspension and formulation stability
- Effective for large and complex molecular structures

Efficacy-Enhancing Adjuvants (NF, DA, TDA Series)

- Maximizes penetration into plant cuticles
- Globally essential adjuvants

Penetrants & Wetting Agents (LA & TDA Series)

- Dramatically reduces surface tension of water
- Ensures complete leaf surface spreading

High-Efficiency Emulsifiers (CO & PE Series)

- Natural castor oil-based (CO)
- EO/PO block emulsifiers/dispersants (PE)

EOD / SEGMENT 4

Industrial Lubricants & Functional Fluids

We provide synthetic PAG lubricants optimized for extreme operating conditions and compatible with next-generation refrigerants.

Standard PAG lubricants (BR / BP Series)

- Water-soluble and oil-soluble industrial gear oils
- Compliant with various global specifications
- High-Viscosity PAG (up to **19,000 cSt**)

Emulsifiers & Wetting Agents (NF / OA / OP Series)

- Rapid substrate wetting for metal cleaning
- Wide HLB range for formulation flexibility

Specialty lubricants

- Di-capped PAG
  - Superior chemical stability and low hygroscopicity
  - Optimized for automotive A/C and high-efficiency compressors
- POE(Polyolester Synthetic Oils)
  - High-stability polyol ester lubricants for industrial and residential refrigeration

Ethanolamine

- pH control & Buffering
- Corrosion Protection
- Emulsion Stability & Dispersibility

ACRYLATE MONOMERS

AM Product Overview

Monofunctional Acrylates
- Reactive diluents, adhesion, and improved wettability
- For prism sheets and adhesives

Difunctional Acrylates
- Balanced crosslink density and flexibility
- For hard coatings, UV inks, and optical films

Tri & Multifunctional Acrylates
- High crosslink density, good hardness, and solvent resistance
- For hard coatings and UV inks

Methacrylates
- Superior weather resistance and less skin irritation than acrylates
- For optical coatings, protective coatings, and dental resins

High/Low RI acrylates
- High refractive index (1.58–1.62) and low refractive index (1.33–1.44)
- For optical films and lenses

Oligomers
- Low VOC, flexibility, adhesion, hardness, and chemical resistance
- For coatings, UV inks, and adhesives

AM Application Markets

UV Coatings
- High durability, scratch resistance, and chemical resistance
- Used in hard coatings, protective coatings, and functional surface layers

UV Printing Inks
- Fast cure speed with low shrinkage
- Used in UV offset, flexography, gravure, screen, inkjet, 3D printing

Adhesives & Sealants
- Good adhesion, flexibility, and fast curing with low VOC content
- Used in electronic assembly, optical bonding, and industrial sealing

Electronics & Optical Materials
- High and low RI monomers for light diffusion and anti-reflection
- Used in anti-reflection and optical control layers for advanced displays

Cosmetic Products
- Low irritation, low shrinkage, and moderate functionality
- Used in UV gel polish

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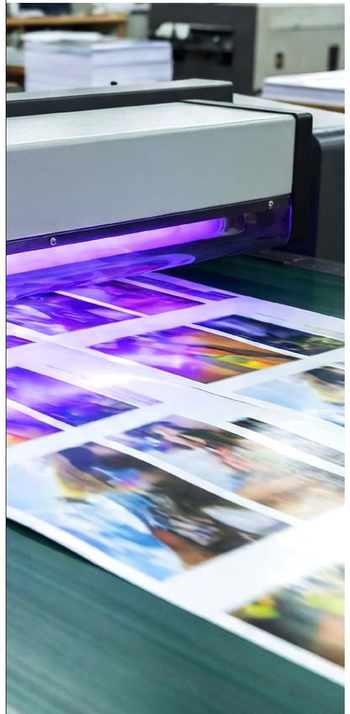
ACRYLATE MONOMERS



High Performance Products

Green Chemical AM R&D delivers innovative specialty monomers for advanced optical, display & eco-friendly coating applications.

High Purity Monomers
- Products: BZMA, THFMA, DCPA, EGDMA
- Purity >98% (GC), Low color index
- For optical films & electronic systems

Bio-based Monomers
- Products: SA, ISB DA, C12~C18 series
- Bio content up to 86% (plant-derived)
- For Eco-friendly UV-cure solutions

High Refractive Index Monomers
- Products: OPPA, BPMA, NTEA
- High refractive index of 1.582 or higher
- For prism sheet for light diffusion purposes

Low-k Dielectric Monomers
- Products: C13~C24 acrylates/methacrylates
- Dielectric constant 2.7 ~ 4.1 (1kHz)
- For electronics & semiconductor use

Low Refractive Index Monomers & Oligomers
- Products: S series, LF series, UA series
- Low refractive index of 1.449 or lower
- For optical films for anti reflective film

Polythiol Monomers
- Products: EGMP, TMMP, TEMPIC, PEMP, DPMP
- Advanced curing control & enhanced adhesion
- For systems requiring improved curing, adhesion

Green Chemical Co., Ltd.

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